SoK: The Quantifier Gap Between Certified Robustness and Airworthiness Assurance

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***Abstract*—Certified-robustness research and aviation assurance both promise guarantees about learned systems, and regulators now ask them to meet. The European Union Aviation Safety Agency (EASA) expects quantifiable evidence about machine-learning constituents. The Specific Operations Risk Assessment (SORA) states its safety objectives per flight hour, and the forthcoming ED-324 standard will codify learning assurance. Meanwhile the robustness literature produces certificates in quantity. We argue that the two rarely meet because they quantify over different things. A certified radius speaks about every perturbation of *one input*. A contingency volume speaks about *one operation*. A per-flight-hour objective speaks about a *population of operations*.**

We systematize this gap. We give every guarantee and every objective a two-axis quantifier type, *scope* times *modality*, and formalize the arguments that connect them as *lifts*: typed inference rules, each of which emits an evidence obligation. A computed lift graph decides, for every pair of guarantee class and objective, whether the guarantee discharges the objective, needs lifts whose assumptions must themselves be evidenced, or is a category error. Four structural results follow. No chain of proofs alone reaches a probability objective: every path crosses at least one statistical obligation. A per-input error budget converts to a per-hour rate without any independence assumption, but temporal confirmation earns its usual exponential credit only under independence. SORA's test-hour ladder is exactly the zero-failure "rule of three". Runtime assurance discharges per-operation objectives, but it moves the gap onto its monitor rather than closing it. A pilot coding of 86 guarantees against 22 objectives exercises the protocol. Only 17% of the lifted guarantees evidence the assumption their lift needs, and no guarantee class discharges a per-flight-hour objective without a lift. A SORA Annex A evidence generator demonstrates one sound path end to end and lists every assumption as an open obligation. Nothing here certifies an aircraft: we map mathematical guarantees onto an evidence framework, and we say where the map has holes.

***Index Terms*—certified robustness, airworthiness, learning assurance, runtime assurance, UAV, SORA, systematization of knowledge**

I. Introduction

An unmanned aerial vehicle (UAV) that navigates with a learned component now meets two kinds of guarantee. The first comes from the certified-robustness and formal-verification literature. It states that a neural network (NN) does not change its decision for any input within a radius of a given point [1]–[3]. It may also state that a closed loop with an NN controller stays inside a reachable tube [4], [5]. The second comes from aviation. It states that an operation keeps the aircraft inside its operational volume with a probability of failure below 10^{-3} or 10^{-4} per flight hour (FH) [6]. It also states that a learned constituent generalizes over its operational design domain (ODD) [7], [8]. The two communities increasingly cite each other. EASA's concept paper asks for robustness and generalization evidence on machine-learning (ML) constituents [7]. The joint standard ED-324/ARP6983 of the European Organisation for Civil Aviation Equipment (EUROCAE) and SAE International will turn that request into objectives [9]. The verification community, for its part, motivates its benchmarks with airborne collision avoidance (ACAS Xu) and aircraft taxiing [2], [10].

The observation: Citation is not transfer. The two kinds of statement rarely support one another, because they quantify over different objects. A certified radius is a statement about *every perturbation of one input*. A reachable tube is a statement about *every disturbance along one trajectory over a horizon*. A contingency volume is a statement about *one operation*. A safety objective per flight hour is a statement about *a population of operations*. The gap is not only in scope but also in modality. "For all" and "with probability at least $1-\delta$" are different claims. So are "with probability over the certifier's own random noise" and "with probability over the operations the aircraft will fly", and the literature routinely conflates them. Evidence transfers from a guarantee to an objective only when the guarantee's quantifier covers the objective's. Otherwise an argument must bridge the two, such as a dynamics bound, a concentration inequality or a runtime monitor. We call such an argument a *lift*. A lift has a price: an assumption that itself needs evidence.

This paper: We systematize the certified-guarantee literature for learned perception, estimation and control by what each guarantee quantifies over. We code aviation and UAS assurance objectives the same way, and we compute which transfers are sound, which need which lifts, and which are category errors. Our contributions are as follows.

- **A quantifier type system and a lift calculus** (Section III). Every guarantee and objective receives a type $\langle \text{scope}, \text{modality} \rangle$. Seven lifts are typed edges between types, and each lift emits named obligations with an evidence kind. Dominance is a partial order (Proposition 1), and transfer along a path is sound with a union-bound confidence (Theorem 1).
- **Four structural results.** No modality comes for free: every path from a "for all" guarantee to a probability objective crosses a statistical obligation (Proposition 2). The rate lift needs no independence assumption, but confirmation credit does (Proposition 3). The SORA 2.5 Annex E test-hour

ladder is the zero-failure rule of three, and testing per frame buys no economy over testing per hour (Proposition 4). Runtime assurance relocates the gap to its monitor rather than closing it (Proposition 5).

- **A registered systematization protocol** (Section IV). It specifies block search strings, two inclusion tiers, a snowballing stop rule, an eight-field codebook, per-field reliability with Cohen's κ, Krippendorff's α, Gwet's AC1 and the prevalence-adjusted bias-adjusted kappa (PABAK), and a large language model (LLM) used as a logged third coder that may never exclude a paper on its own.
- **A computed transfer table and a pilot study** (Sections VII and IX). On a pilot of 86 guarantees in 17 classes against 22 objectives, the table has 102 cells: 8 discharges, 66 lifts and 28 category errors. No class discharges a per-flight-hour objective without a lift. The pilot exercises every step of the protocol. It does not estimate the field, and we say so wherever it matters.
- **An evidence generator** (Section VIII). From the committed results of a published composition certificate [11], it emits a SORA Annex A-shaped evidence pack in which every lift's assumption appears as an obligation with its status.

Scope and honesty: Three boundaries belong with these claims. First, the transfer table is computed from a calculus that we define. Its verdicts are only as good as the lift rules, which we therefore state in full (Table II) and release as code. Second, the corpus figures we report come from a pilot. Both pilot coders were LLM instances working independently from the codebook, so the reliability we report measures the clarity of the codebook, not agreement between experts. The registered full study replaces every pilot number without hand editing, because each number is a macro generated from result files. Third, nothing in this paper certifies an aircraft. We map mathematical guarantees onto an evidence framework. Several regulatory documents we code were in flux at our cut-off date (7 October 2026), and we key our claims to stable concepts rather than clause numbers.

II. Preliminaries and Problem Statement

A. Two Vocabularies

Certified guarantees: A *certified guarantee* about a learned component f is a sound statement, or a statistically valid one, of the form "for every x in a set S, property $\varphi(f,x)$ holds". It may also take the form "with probability at least $1-\delta$ over a stated distribution, φ holds". Robustness certificates set $S=\{x' : \|x'-x_0\| \le r\}$ and φ to "the decision does not change". They are computed by complete solvers based on satisfiability modulo theories (SMT) or mixed-integer programming [2], [12]. They are also computed by sound bound propagation, as in interval bound propagation (IBP) and CROWN [3], [13], [14], by abstract interpretation [15], [16], and by randomized smoothing (RS), whose certificate holds with confidence $1-\alpha$ over the certifier's own Monte Carlo noise [1]. Closed-loop verification of NN control systems (NNCS) sets S to an initial-state set and φ to "the trajectory stays in a safe set over horizon T" [4], [5], [17].

Assurance objectives: An *assurance objective* is a requirement that an applicant must show is met. SORA 2.5 of the Joint Authorities for Rulemaking of Unmanned Systems (JARUS) sets a target level of safety (TLS) of fewer than 10^{-6} fatalities per flight hour [18]. Its Annex E requires the probability of leaving the operational volume to be below 10^{-3}/FH for low and medium robustness and below 10^{-4}/FH for high robustness [6]. Its Annex A defines the operational volume as the flight geography plus a *contingency volume*, sized from navigation accuracy, position-holding error, reaction distance and the contingency manoeuvre [19]. Manned-aircraft system-safety guidance states catastrophic failure conditions per flight hour, down to 10^{-9} [20]. EASA's learning assurance asks for generalization bounds over the ODD and for robustness of the model [7]. The ASTM F3269 practice for run-time assurance (RTA) bounds a complex function's behaviour on every execution with a monitor and a recovery function [21].

B. The Running Example: A Composition Certificate

A recent composition [11] bounds the mission-completion rate (MCR) of a UAV against an adversary who delays relayed navigation messages just below an intrusion detector's threshold θ. Its chain is

$$\theta \mapsto \Delta(\theta) \mapsto \delta = \gamma\Delta \mapsto \rho = \delta\, e^{LT_c} \le m \mapsto \mathrm{MCR} \ge f(\delta), \tag{1}$$

where $\Delta(\theta) \le H\min(\theta, s)$ is the hidden delay over H compromised hops with contact-window slack s. Here γ is a measured delay-to-position rate, L a Lipschitz constant of the navigation dynamics, T_c a re-anchoring horizon of the extended Kalman filter (EKF), and m a corridor margin. A follow-on paper [22] treats Δ and γ as design variables traded through authentication. The parent's three stated conditions, on dynamics, re-anchoring and interface transfer, are exactly the assumptions of the lifts in Eq. (1). That is why it serves as our running example rather than our subject.

C. Problem Statement

Let $\mathscr{G}$ be the set of certified guarantees about learned components of autonomous vehicles published from 2017 to 2026, and let $\mathscr{J}$ be the set of assurance objectives stated by aviation and UAS regulation and standards at a fixed cut-off date. For a guarantee $g \in \mathscr{G}$ and an objective $o \in \mathscr{J}$ we ask whether g *discharges* o, whether g is *evidence for* o only through stated lifts, or whether citing g for o is a *category error*. We further ask which assumptions the lifts introduce, and whether the literature accounts for them. We test four hypotheses (H), each of which could fail.

H1 Most certified guarantees for learned *perception* quantify per input, and therefore discharge no assurance objective without a lift. We expect H1 to fail for learned *control*, where closed-loop reachability is mature.

H2 Lifts are used, but their assumptions are rarely accounted for as obligations that need evidence.

TABLE I
NOTATION.

$\tau = \langle \sigma, \mu \rangle$	quantifier type: scope σ, modality μ
$\mathsf{in}, \mathsf{tr}, \mathsf{op}$	one input; trajectories over a horizon; one operation
$\mathsf{pop_d}, \mathsf{pop_h}$	population per demand; rate per flight hour
$\forall, \forall_\alpha$	for all; for all with certifier confidence $1-\alpha$
$\mathbb{P}, \mathbb{E}, \exists$	probability over the environment; expectation; exists
ℓ, $\mathscr{O}(\ell)$	lift; its set of obligations
π, $\mathscr{O}(\pi)$	path of lifts; its obligations (assurance deficit)
δ_o, ε_o	failure probability of evidence for o; its error term
$\succeq$	dominance (obligation-free transfer)
f, p, λ	decision rate (Hz); per-decision hazard probability; rate per hour
k, ρ	confirmation length; error persistence $\Pr(e_t \mid e_{t-1})$
$\Delta, \gamma, L, T_c, m$	parent certificate: hidden delay, rate, Lipschitz constant, horizon, margin

H3 Per-flight-hour and ODD objectives are natural targets for statistical guarantees, yet statistical guarantees for learned perception are the least developed class.

H4 Runtime assurance is the dominant practical bridge for objectives that no offline certificate discharges.

Table I collects the notation. Fig. 1 shows how the answers are assembled.

III. Methodology I: A Quantifier Type System

The framework (Fig. 1) has five stages. This section develops stage (3), the theory against which the corpus is coded. Section IV develops stages (1), (2) and (5). Proofs omitted here are in Appendix A.

A. Types

Definition 1 (Quantifier type)**.** A guarantee or objective has type $\tau = \langle \sigma, \mu \rangle$. The *scope* $\sigma \in \{\mathsf{in}, \mathsf{tr}, \mathsf{op}, \mathsf{pop_d}, \mathsf{pop_h}\}$ names the object quantified over: inputs in a neighbourhood or region of one point (in), trajectories over a bounded horizon (tr), one complete operation or every execution (op), a distribution of demands such as inputs, environments or operations ($\mathsf{pop_d}$), or a rate per flight hour ($\mathsf{pop_h}$). The *modality* $\mu \in \{\forall, \forall_\alpha, \mathbb{P}, \mathbb{E}, \exists\}$ names the kind of claim: sound for all ($\forall$), for all with confidence $1-\alpha$ over the certifier's own randomness ($\forall_\alpha$), a probability bound over the environment or data ($\mathbb{P}$), a bound on an expectation ($\mathbb{E}$), or an existential witness ($\exists$).

Two distinctions in Definition 1 carry most of the weight. First, $\forall_\alpha$ is not $\mathbb{P}$. The confidence of randomized smoothing is over the certifier's coins. It says nothing about how often the aircraft meets an input outside the certified ball. Second, $\mathsf{pop_d}$ is not $\mathsf{pop_h}$. A misclassification probability per image, or a failure probability per mission, is a statement per demand. A target per flight hour also needs an exposure model. Fig. 2 draws the type plane.

B. Dominance

Definition 2 (Dominance)**.** Type τ *dominates* τ', written $\tau \succeq \tau'$, if a statement of type τ entails a statement of type τ' about the same property with no further evidence. A guarantee g *discharges* an objective o if $\tau_g \succeq \tau_o$ and the perturbation model of g covers the hazard of o.

Dominance is generated by two obligation-free rules. A probability bound on a failure event is a bound on the expectation of its indicator, so $\langle \sigma, \mathbb{P} \rangle \succeq \langle \sigma, \mathbb{E} \rangle$ for population scopes. A probability bound per operation is a statement about the population of operations, so $\langle \mathsf{op}, \mathbb{P} \rangle \succeq \langle \mathsf{pop_d}, \mathbb{P} \rangle$.

Proposition 1 (Dominance is a partial order)**.** *$\succeq$ is reflexive, transitive and antisymmetric. Under it, $\langle \mathsf{in}, \forall \rangle$ and $\langle \mathsf{pop_h}, \mathbb{P} \rangle$ are incomparable, and so are $\langle \mathsf{op}, \forall \rangle$ and $\langle \mathsf{pop_d}, \mathbb{P} \rangle$.*

The proposition is the formal content of the gap. Under $\succeq$, a radius is not a weaker or stronger kind of per-flight-hour claim. It is a different claim, and moving between the two is an argument, not a weakening. The second incomparability surprises at first: a per-operation "for all" looks stronger than any probability. It is stronger only on its certified set. To become a probability over deployed operations, it needs evidence that deployed operations stay in that set.

C. Lifts and Obligations

Definition 3 (Lift)**.** A *lift* $\ell : \tau \to \tau'$ is an inference rule whose premise is a statement of type τ and whose conclusion is a statement of type τ'. It is valid only together with a set of side conditions $\mathscr{O}(\ell)$, its *obligations*. Each obligation has an evidence kind: *proof*, *model*, *statistical*, or *architectural*.

Table II lists the seven lifts of the calculus. Each one corresponds to a family of arguments in the corpus. $\mathsf{L_{dyn}}$ is the Lipschitz–Grönwall step of the running example [23] and the reachability step of NNCS verification. $\mathsf{L_{op}}$ is the operational-profile argument of reliability assessment [24], [25]. $\mathsf{L_{sup}}$ is the argument, usually left implicit, that deployed operations stay inside the certified set. $\mathsf{L_{mon}}$ is the Simplex argument [26], [27].

D. Transfer Paths and Soundness

Definition 4 (Transfer path and assurance deficit)**.** A *transfer path* from guarantee g to objective o is a sequence of lifts $\pi = \ell_1 \cdots \ell_k$ with ℓ_1 starting at τ_g and ℓ_k ending at a type that dominates τ_o. Its *assurance deficit* is the obligation set $\mathscr{O}(\pi) = \bigcup_i \mathscr{O}(\ell_i)$, extended by $\mathsf{O_{thr}}$ when the hazard is not covered. The *verdict* is **D** (discharge) if some path has an empty deficit, $\mathbf{L}_k$ if the least deficit has k obligations, and $\times$ (category error) if no path exists.

Theorem 1 (Transfer soundness)**.** *Let g hold with confidence at least $1-\delta_0$, where $\delta_0 = 0$ for $\forall$ and $\delta_0 = \alpha$ for $\forall_\alpha$. Let π be a transfer path from g to o, and let each obligation $\mathsf{O} \in \mathscr{O}(\pi)$ be discharged by evidence that is valid with probability at least $1-\delta_{\mathsf{O}}$, where $\delta_{\mathsf{O}} = 0$ for proofs. Then the conclusion of π holds with confidence at least*

$$1-\delta_\pi, \qquad \delta_\pi \;=\; \delta_0 + \sum_{\mathsf{O} \in \mathscr{O}(\pi)} \delta_{\mathsf{O}}, \tag{2}$$

and its error term composes along the rules of Table II, additively through $\mathsf{L_{op}}$ and $\mathsf{L_{sup}}$ and multiplicatively by n_d through $\mathsf{L_{rate}}$. If the conclusion's type dominates τ_o, the objective's statement holds at that confidence.

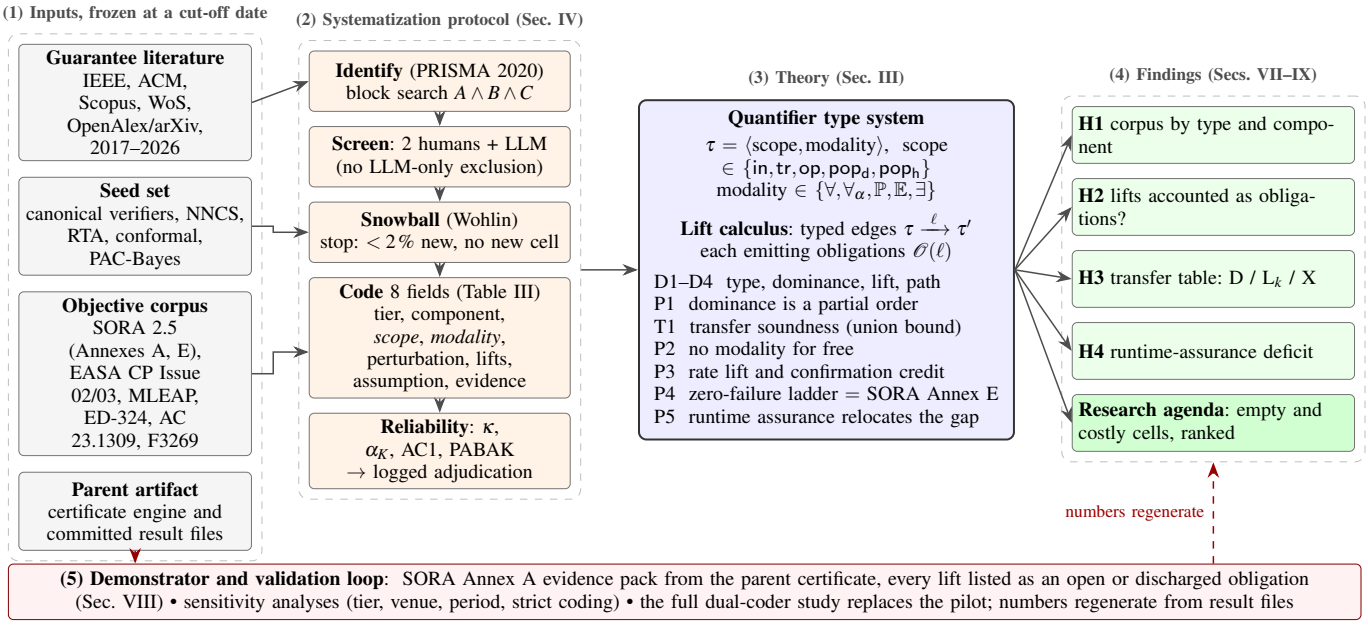


Fig. 1. Methodological framework. (1) Two corpora enter the study: guarantees about learned components, and assurance objectives from aviation regulation and standards, both frozen at a cut-off date. (2) A registered protocol identifies, screens, snowballs and codes the guarantees. Two coders work independently, a language model acts as a logged third coder, and reliability is reported per field before adjudication. (3) Every guarantee and objective receives a quantifier type, and a calculus of typed lifts computes, for each pair, whether the guarantee discharges the objective, needs lifts whose assumptions become obligations, or is a category error. (4) The coded corpus and the calculus answer H1–H4 and yield a ranked research agenda. (5) An evidence generator instantiates one sound path end to end, and the validation loop regenerates every number from result files.

Theorem 1 is elementary, a union bound over the events on which some premise fails. Its use is accounting. A paper that lifts a guarantee without naming δ_O for each obligation has not established Eq. (2). It has established a conditional statement whose conditions an assurance case must carry. This is H2 made measurable: the coding scheme records, per guarantee, whether the assumption of each lift is unstated, stated, stated and evidenced, or evidenced with a confidence that enters the final claim.

E. Four Structural Consequences

Proposition 2 (No modality for free)**.** *Every transfer path from a type with modality $\forall$ or $\forall_\alpha$ to a type with modality $\mathbb{P}$ contains at least one lift with a statistical obligation.*

The proposition states something the field often argues informally [28], [29], and the calculus makes it exact. No chain of proofs, however sound or tight, turns a radius into a probability over deployment. At some point an argument must sample the world the aircraft will fly in. Better certificates shrink the other terms of Eq. (2). They cannot remove the statistical one.

Proposition 3 (Rate lift and confirmation credit)**.** *Let a component take f decisions per second, each hazardous with stationary marginal probability at most p. (a) The expected number of hazardous decisions per flight hour is at most $3600fp$, whatever the dependence between decisions. Hence a per-hour target λ is met whenever $p \le p^\star = \lambda/(3600f)$. (b) Suppose a hazard requires $k \ge 2$ consecutive erroneous decisions. Under arbitrary stationary dependence, the rate of hazardous runs is at most $3600fp/k$. Under a two-state Markov error process with persistence $\rho = \Pr(e_t \mid e_{t-1})$, it equals $3600fp(1-\rho)\rho^{k-1} \le 3600fp\,(k-1)^{k-1}/k^k$. Under independence it is approximately $3600fp^k$.*

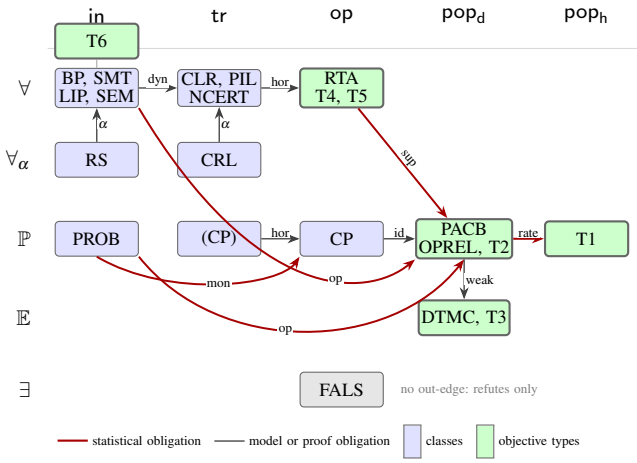


Fig. 2. The type plane. Columns are scopes, rows are modalities, and boxes place the pilot's guarantee classes and objective types T1–T6. Each arrow is a lift with obligations (Table II). Red arrows carry a statistical obligation, and every path from a $\forall$ state to a $\mathbb{P}$ state crosses one (Proposition 2). No arrow points left, and none leaves $\exists$ or $\mathbb{E}$.

At 30 Hz and 10^{-4}/FH, part (a) gives $p^\star = 9.3 \times 10^{-10}$ per frame (108,000 decisions per hour). At 10^{-9}/FH it gives $p^\star = 9.3 \times 10^{-15}$. Part (b) explains a common move. Temporal

TABLE II

THE LIFT CALCULUS. EACH LIFT IS AN INFERENCE RULE FROM A PREMISE TYPE TO A CONCLUSION TYPE, VALID TOGETHER WITH ITS OBLIGATIONS. KIND: P PROOF, M MODEL, S STATISTICAL, A ARCHITECTURAL. THE RULES ARE IMPLEMENTED AS THE EDGE SET OF `QUANTIFIER.PY`, AND THE PROPOSITIONS ARE CHECKED ON IT MECHANICALLY.

Lift	Edge	Rule (premise $\Rightarrow$ conclusion)	Obligations	Kind
L_{α}	$\langle \sigma, \forall_{\alpha} \rangle \to \langle \sigma, \forall \rangle$	certificate valid with probability $1-\alpha$ over the certifier's coins $\Rightarrow$ certificate valid, at confidence $1-\alpha$	O_{α}: accept α	P
L_{dyn}	$\langle \mathsf{in}, \forall \rangle \to \langle \mathsf{tr}, \forall \rangle$	$\|\delta u(t)\| \le r$ for all t, drift L-Lipschitz $\Rightarrow$ $\|e(t)\| \le r(e^{Lt}-1)$ on $[0, T_c]$	O_{lip}: L and model error over the operating region	M
L_{hor}	$\langle \mathsf{tr}, \forall \rangle \to \langle \mathsf{op}, \forall \rangle$; $\langle \mathsf{tr}, \mathbb{P} \rangle \to \langle \mathsf{op}, \mathbb{P} \rangle$	tube bound per window, re-anchoring every T_c $\Rightarrow$ bound on $\sup_{[0,T]} \|e\|$	O_{hor}: re-anchoring or horizon covers the operation	M
L_{op}	$\langle \mathsf{in}, \forall \mid \mathbb{P} \rangle \to \langle \mathsf{pop}_{\mathsf{d}}, \mathbb{P} \rangle$	robust on cells $c \in \mathscr{C} \Rightarrow \Pr_{x \sim \text{OP}}[\text{error}] \le \text{OP}(\mathscr{X} \setminus \cup \mathscr{C}) + \varepsilon_{\text{prof}}$	O_{prof}: operational profile and per-cell robustness	S
L_{sup}	$\langle \mathsf{op}, \forall \rangle \to \langle \mathsf{pop}_{\mathsf{d}}, \mathbb{P} \rangle$	$\varphi(\omega)$ for all $\omega \in S \Rightarrow \Pr_{\omega \sim D}[\neg\varphi] \le \Pr_D[\omega \notin S] \le \varepsilon_{\text{odd}}$	O_{odd}: deployment stays in the certified set	S
L_{rate}	$\langle \mathsf{pop}_{\mathsf{d}}, \mathbb{P} \rangle \to \langle \mathsf{pop}_{\mathsf{h}}, \mathbb{P} \rangle$	per-demand hazard $\le p$, n_d demands per hour $\Rightarrow$ $\lambda \le n_d p$ (Prop. 3)	O_{dep}: exposure n_d, stationarity; dependence model for confirmation credit	S
L_{mon}	$\langle \mathsf{in}, \forall \mid \mathbb{P} \rangle \to \langle \mathsf{op}, \mathbb{P} \rangle$	monitor flags hazardous inputs, switch within latency, verified recovery $\Rightarrow$ per-operation safety except on monitor misses	O_{mcov}, O_{lat}, O_{rec}, O_{ind}	S,M,P,A

Free edges (no obligation): $\langle \mathsf{op}, \mathbb{P} \rangle \to \langle \mathsf{pop}_{\mathsf{d}}, \mathbb{P} \rangle$ (identity of reading); $\langle \mathsf{pop}_{\cdot}, \mathbb{P} \rangle \to \langle \mathsf{pop}_{\cdot}, \mathbb{E} \rangle$ (indicator expectation).
Hazard coverage: O_{thr} is added whenever the guarantee's perturbation model does not cover the objective's hazard (natural, adversarial, or both).

filtering is credited with turning a per-frame error of 9.7×10^{-4} into a per-hour rate of 10^{-4}, a gain of about 1×10^{6}. That credit is real only under independence. A first-order Markov process guarantees a gain of only $k^k/(k-1)^{k-1} = 6.75$ for $k = 3$, and arbitrary dependence guarantees only k (Fig. 4(b)). The obligation O_{dep} is therefore where perception-based per-hour arguments stand or fall.

Proposition 4 (Zero-failure ladder)**.** *(a) Under a homogeneous Poisson failure process, T failure-free hours reject $\lambda \ge \lambda_0$ at level a if and only if $T \ge \ln(1/a)/\lambda_0$. At $a = 0.05$ the SORA 2.5 Annex E test durations of 30, 300, 3000 and 30 000 h [6] correspond to $\lambda_0 = 10^{-1}, \ldots, 10^{-4}$/FH within 0.15 %. (b) Demonstrating $p \le \lambda_0/(3600 f)$ per frame by zero-failure testing requires exactly as many hours of data as (a). It also replaces an independence assumption between hours with one between frames, which is harder to defend.*

Part (a) shows that a regulator already accepts a concentration lift, the "rule of three" [30], with explicit arithmetic. We use it as the worked example of a regulator-endorsed $\langle \mathsf{pop}_{\mathsf{d}}, \mathbb{P} \rangle \to \langle \mathsf{pop}_{\mathsf{h}}, \mathbb{P} \rangle$ step. Part (b) closes a tempting shortcut: 10^{-4}/FH at 30 Hz needs 3.2×10^{9} failure-free frames, or 29,957 h, and 10^{-7}/FH needs about 3,417 years of continuous flight [31], [32].

Proposition 5 (Runtime assurance relocates the gap)**.** *In the calculus, $\langle \mathsf{op}, \forall \rangle$ objectives are discharged directly only by a guarantee of type $\langle \mathsf{op}, \forall \rangle$. Such a guarantee from runtime assurance is conditional on $\{\mathsf{O}_{\text{mcov}}, \mathsf{O}_{\text{lat}}, \mathsf{O}_{\text{rec}}, \mathsf{O}_{\text{ind}}\}$. When the monitor's decision depends on a learned perception output, O_{mcov} is a statement of type $\langle \mathsf{pop}_{\mathsf{d}}, \mathbb{P} \rangle$ about that output, the same type as the gap the architecture was introduced to bridge.*

Corollary 1 (The cheapest route skips the operation)**.** *From $\langle \mathsf{in}, \forall \rangle$, the least-obligation route to $\langle \mathsf{pop}_{\mathsf{h}}, \mathbb{P} \rangle$ is $\mathsf{L}_{\text{op}}\mathsf{L}_{\text{rate}}$, with two obligations. It never visits $\langle \mathsf{op}, \forall \rangle$. Every route that also yields a per-operation guarantee costs at least four obligations: O_{lip}, O_{hor}, O_{odd} and O_{dep}.*

The corollary separates two things an applicant may want. One is a per-hour number. The other is a per-operation envelope, which is what sizes a contingency volume. The cheaper route yields the first without the second. The running example takes the longer route and pays for it, and Section VIII shows what it buys.

IV. METHODOLOGY II: SYSTEMATIZATION PROTOCOL

A. The Guarantee Corpus

Sources and search: The protocol follows the systematic-review guidelines of Kitchenham and Charters [33] and reports with the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) 2020 [34]. We search OpenAlex [35] for 2017–2026, retrieving every result of every query. OpenAlex indexes the Crossref records of publications in IEEE Xplore and the ACM Digital Library, together with arXiv and DBLP, so one search reaches most of what those libraries hold; we did not run separate searches of the subscription databases Scopus and Web of Science. Each query conjoins three blocks: A, a guarantee term (16 phrases, from "certified robustness" and "reachability analysis" to "conformal prediction", "PAC-Bayes" and "runtime assurance"); B, a learning term; and C, an autonomy term (Appendix B). Through OpenAlex we also sweep the complete 2017–2026 tables of contents of IEEE S&P, USENIX Security, CCS, NDSS, CAV, TACAS, HSCC, ICCPS, EMSOFT, NFM, SAFECOMP, DSN, NeurIPS, ICML, ICLR, CoRL, CDC and DASC, keeping papers whose title contains a block-A phrase. Records are merged when they share a DOI, or a title with publication years at most two apart, so a preprint and its published version count once. Every request is logged with a timestamp,

TABLE III
CODEBOOK (VERSION 1.1). ONE ROW PER GUARANTEE.

Field	Values
tier	A (vehicle task) / B (method anchor)
component	perception / estimation / control / planning / end-to-end
scope	in / tr / op / pop
modality	$\forall$ / $\forall_\alpha$ / $\mathbb{P}$ / $\mathbb{E}$ / $\exists$
perturbation	norm / semantic / patch / disturbance / distribution / poisoning / none
lifts (list)	L_{dyn}, L_{hor}, L_{gen}, L_{op}, L_{rate}, L_{conc}, L_{sup}, L_{mon}, none
assumption	U unstated / S stated / SE stated and evidenced / SEC evidenced with a confidence that enters the claim
evidence	proof / measured / simulated / modelled

and we report the search's sensitivity against the pilot set (Section IV-F): the share of its papers the search retrieves.

Inclusion: We include a paper if it states a formal, statistical or architectural guarantee about a learned component, or about a system containing one, and if the object its guarantee quantifies over can be identified. We exclude empirical attack studies without a guarantee, guarantees about non-learned components, and theses that duplicate a journal version.

Two tiers: The inclusion rule in our proposal required evaluation on an autonomous vehicle. Applied literally, it would exclude the canonical verifiers that define the guarantee classes, which are evaluated on image benchmarks. We therefore code two tiers. *Tier A* contains guarantees evaluated on a vehicle task, including standard vehicle benchmarks such as ACAS Xu, TaxiNet, adaptive cruise control and quadrotors. *Tier B* contains method anchors evaluated only on generic benchmarks. Every statistic is reported for tier A and for both tiers together, so that a reader can see how much of a finding is carried by the anchors.

Snowballing: We snowball backward and forward from the included set, following Wohlin [36]. We stop when two consecutive iterations each add fewer than 2 % new inclusions *and* no iteration adds a new cell of the type plane. The second condition matters more: an SoK can miss papers, but it should not miss a kind of guarantee.

B. Codebook

The unit of coding is a guarantee, the headline claim of a paper. Table III lists the eight fields. Each uses a closed vocabulary. Three decision rules prevent the confusions that Section III warns against. Randomized-smoothing confidence is coded $\forall_\alpha$, never $\mathbb{P}$. Falsification is coded $\exists$, because it can only refute. A certificate checked by sampling over its domain is coded with L_{conc}, because sampling has entered the claim. The pilot added three clarifying rules, which froze the codebook at version 1.1 before the full study (Appendix C).

Two codebook lifts are not edges of the calculus. L_{gen} (a generative, abstraction or contract surrogate for the real sensor) and L_{conc} (a concentration, PAC, conformal or scenario argument from samples) change how a guarantee is *obtained*, not its type. They enter the transfer table as intrinsic obligations of the class, O_{cov} and O_{iid}, alongside the obligations of the calculus edges a class's members already use.

C. Coders, Reliability and the LLM Third Coder

Two coders code every paper independently. We report, per field, percent agreement, Cohen's κ [37], Krippendorff's nominal α [38], Gwet's AC1 [39] and PABAK, each with a percentile bootstrap interval over papers. We accept $\alpha \geq 0.80$, and $\alpha \geq 0.667$ for tentative conclusions only. We report AC1 and PABAK because the corpus is expected to be skewed toward in. Under skew, κ and α can be low despite high agreement, the prevalence paradox [40]. The set-valued lifts field is scored per lift, as a binary variable, and by mean Jaccard similarity. Disagreements are adjudicated by discussion, and every change is logged with its rationale.

An LLM acts as a third coder under four rules. It runs at a fixed model version with a registered prompt and codebook. Each item is coded three times, and run-to-run agreement is reported. It may flag a screening exclusion for human review but never exclude alone. Its agreement with the human consensus is reported separately. Published evaluations of LLM screening report moderate-to-substantial agreement that depends strongly on the clarity of the criteria [41], [42]. That dependence is why we use the LLM as an instrument to measure, not as a coder to trust.

D. The Objective Corpus

We code 22 objectives from nine documents: the SORA 2.5 main body and Annexes A and E [6], [18], [19]; EASA's AI concept paper Issue 02 and Proposed Issue 03 [7], [43]; the final report of EASA's Machine Learning Application Approval (MLEAP) project [8]; ED-324/ARP6983 at the level of concepts [9]; the Federal Aviation Administration (FAA) advisory circular AC 23.1309-1E and EASA's acceptable means of compliance AMC 25.1309 [20], [44]; ASTM F3269-21 [21]; and DO-326A [45]. Of these, 14 were checked against the primary text. The others, marked † in Table IV, rest on secondary sources and are flagged for checking. Objectives with no behavioural quantifier, such as the structural-coverage objectives of DO-178C [46], are recorded but excluded from the transfer table: they constrain the process, not the behaviour. The corpus is frozen at 7 October 2026. ED-324 and the Proposed Issue 03 were moving at that date, so we code them only at the level of stable concepts.

E. Computing the Transfer Table

The objectives fall into six types, by type and hazard (Table VI): per flight hour (T1), generalization over the ODD (T2), average error (T3), per operation (T4), security threat condition (T5) and local robustness (T6). Each guarantee class receives the modal type and modal perturbation of its adjudicated members. Its *intrinsic* obligations are those of the lifts that at least half of its members use. For each class and objective type, we compute the least-obligation path with Dijkstra's algorithm over the edge set of Table II and add O_{thr} when the hazard is not covered. Norm-bounded, patch, semantic and poisoning models cover adversarial hazards; disturbance and distribution models cover natural ones. One class-level override is documented: a Simplex guarantee holds

for arbitrary behaviour of the monitored component, but not for attacks on the monitor's own inputs, so it covers natural hazards only.

F. Pilot and Full Study

This paper reports a pilot that exercises every step of the protocol. The pilot corpus has 86 guarantees (46 in tier A, 40 in tier B) in 17 classes. It is a purposive seed set drawn from the canonical systems and from a scoping review, not a sample, so its distributions illustrate the pipeline and do not estimate the field. Two LLM instances coded it independently from the codebook, the second without access to the first's codes. One coder's codes were then adjudicated against the other's with 50 logged changes across 37 papers. The full study, specified as a runbook in the artifact, replaces both LLM coders with human coders, keeps the LLM as a third coder, and regenerates every number in Sections VII–IX.

G. The Evidence Generator

The generator reads a certified configuration of the running example from its committed result files. It emits a SORA Annex A-shaped pack in which: the ODD is the parent's measured operating region; the position-holding term of the contingency volume is the certified tube radius under the stated adversary; every other term of the contingency volume is an open obligation of the operator; and every lift of Eq. (1) appears, typed, with its obligation and status. The last point is the generator's contribution. A compliance pack that lists the certificate but hides its assumptions would commit exactly the category error this paper names.

V. The Guarantee Landscape

We now systematize what each level offers, with the classes of the pilot in parentheses. Appendix D lists class membership.

A. Per Input

Per-input certificates are the most developed class. Complete methods decide properties over input regions exactly [2], [12], [47] (SMT). Incomplete sound methods scale through linear relaxations, branch and bound, and cutting planes [3], [13]–[16], [48], [49] (BP). Lipschitz-constrained architectures make a certified margin a by-product of training [50]–[52] (LIP). Randomized smoothing scales certification to large models, including detection and segmentation [1], [53]–[57] (RS). Semantic, geometric and patch certificates replace the norm ball with a parametrized set of rotations, illumination changes or bounded patches [58]–[64] (SEM). These parametrized sets are the closest the literature comes to EASA's language of an ODD. Each set is still a neighbourhood of one input, not a distribution over the operations the aircraft will fly. The international verification-of-NN competition (VNN-COMP) standardizes this level: its specifications are pre- and postconditions on one network [65].

Aviation case studies sit here too. ACAS Xu properties are checked over regions of encounter geometry [2], [66]. Industrial studies have assessed an exact ACAS Xu implementation against draft ED-324 objectives, with formal verification discharging some learning-assurance objectives [67], [68]. The region quantifier makes these statements per region of state space, which is still per input in our sense. Lifting them to an encounter needs a closed-loop model.

B. Per Trajectory

For learned *control* with low-dimensional state inputs, closed-loop verification is mature (CLR). Taylor-model, star-set, polynomial-zonotope and semidefinite methods compute reachable tubes for NN controllers in feedback with known plants [4], [5], [17], [69]–[78]. The ARCH-COMP category for AI and NN control systems (AINNCS) benchmarks them yearly [79]. Its 2025 report notes that tools abstracting the problem differently produced disagreeing results on two benchmarks, direct evidence that the quantified object itself can differ between tools. Neural Lyapunov and barrier certificates give $\forall$-trajectory statements over a domain [80]–[82] (NCERT). Certified reinforcement learning bounds actions or cumulative reward under bounded observation perturbations [83]–[86] (CRL).

For learned *perception* in the loop, every result rests on a surrogate (PIL). The surrogate is a generative model that maps states to plausible images [10], a learned abstraction [87], a perception contract [88]–[90], a camera model [91], or a composition of these [92]–[94]. The guarantee then covers the surrogate's range, not real images. A probabilistic analysis of Boeing's taxiing system states plainly that it is best viewed as average-case rather than worst-case [95] (DTMC). This is the pattern H2 predicts: a type-changing lift (L_{gen}) whose assumption, surrogate coverage, is of type $\langle \mathsf{pop}_{\mathsf{d}}, \mathbb{P} \rangle$.

C. Per Operation

Runtime assurance gives per-execution guarantees by switching from an unverified controller to a verified fallback when a monitor fires [26], [27], [96]–[99] (RTA). Shields enforce a safety specification on a learning agent's actions [100]. ASTM F3269 standardizes the architecture for aircraft and makes "RTA system coverage" an explicit concern [21]. Monitors for learned perception detect activation patterns or inputs outside the training distribution [101]–[103] (MON). Their detection rates are population statements, usually reported without confidence. Falsification searches mission specifications in signal temporal logic (STL) for counterexamples [104]–[107] (FALS). It is the per-operation counterpart of testing: it can refute, never discharge.

D. Per Population

Statistical guarantees bound generalization by PAC-Bayes [108]–[110], including for policies across novel environments [111] (PACB). Conformal prediction calibrates trajectory predictors and STL runtime verification with distribution-free coverage under exchangeability [112]–[116], and under bounded distribution shift [117], [118] (CP). Probabilistic robustness estimates the mass of a neighbourhood that is misclassified [119]–[121] (PROB). It is a population statement over a ball, not over deployment. Operational-profile

reliability assessment is the one line of work that lifts per-input robustness to a probability of misclassification per demand. It then argues from that to system-level safety targets [24], [25], [122], [123], alongside rare-event simulation of end-to-end systems [124], [125] (OPREL). It is the literature's most explicit $\langle \mathsf{in}, \forall \rangle \to \langle \mathsf{pop_d}, \mathbb{P} \rangle$ lift, and it states its assumptions. Even so, it treats them as modelling choices rather than as obligations with a confidence. The negative results on demonstrating ultra-high reliability by testing bound what any of these can achieve [28], [31], [32].

VI. The Objective Landscape

Table IV codes the objective corpus. Three observations stand out.

Regulators quantify per flight hour, and they mix quantifiers inside one document. SORA's target levels of safety and containment criteria are $\langle \mathsf{pop_h}, \mathbb{P} \rangle$. Its operational and contingency volumes are $\langle \mathsf{op}, \forall \rangle$ geometry, and it states that the specific assurance and integrity level (SAIL) is not quantitative [18]. EASA asks for generalization bounds over the ODD ($\langle \mathsf{pop_d}, \mathbb{P} \rangle$) and robustness of the model ($\langle \mathsf{in}, \forall \rangle$) in one learning-assurance process. In the same document, it notes that generalization-based error estimates are quantities on average ($\langle \mathsf{pop_d}, \mathbb{E} \rangle$) [7].

The ODD defines the domain of the population quantifier. EASA distinguishes an operational domain (OD) at system level from an ODD at constituent level, the latter a refinement of the former [7]. Under our types, the ODD is the support of D in $\mathsf{L_{sup}}$ and the domain of OP in $\mathsf{L_{op}}$. Data completeness and representativeness objectives are therefore the evidence that discharges $\mathsf{O_{odd}}$ and $\mathsf{O_{prof}}$ [8].

Architectural mitigation is the regulators' own bridge. SORA's "no single failure" criterion, AC 23.1309's equivalent, ASTM F3269, and EASA's monitoring of inputs outside the ODD are all $\langle \mathsf{op}, \forall \rangle$ objectives that runtime assurance targets directly [6], [7], [20], [21]. Secondary commentary on the Proposed Issue 03 reports an independent operational oversight system for higher automation levels [43]. We record that as $\langle \mathsf{op}, \forall \rangle$ and flag it for checking against the primary text.

VII. Results

A. Reliability of the Pilot Coding

Table V reports agreement between the two independent pilot coders. The fields the findings rest on are reliable. For the type, scope times modality, Krippendorff's α is 0.83 (95 % bootstrap interval $[0.74, 0.91]$). For modality alone it is 0.88. Two fields fall short, and both teach something. Perturbation ($\alpha = 0.69$) failed on one ambiguity, whether initial-state reachability without a disturbance input is "none" or "disturbance". Rule CLR-1 of codebook version 1.1 resolves it. The assumption field shows the prevalence paradox: 85 % agreement but $\kappa = 0.47$, because "stated" dominates. AC1 is 0.83. Its lower interval bound of 0.21 marks it as the field the full study must watch. The lifts field has mean Jaccard similarity 0.90.

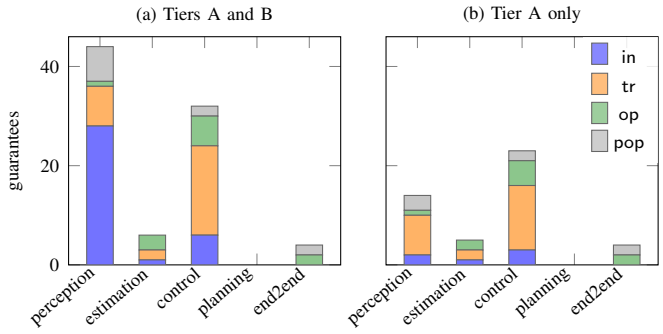


Fig. 3. The pilot corpus by component and scope after adjudication. (a) Across both tiers, perception guarantees are mostly per input and control guarantees mostly per trajectory. (b) Among guarantees evaluated on vehicles (tier A), perception is dominated by closed-loop verification through surrogates. The pilot was seeded purposively, so these shares illustrate the pipeline and do not estimate the field (Sec. XI). Provenance: PILOT CODING.

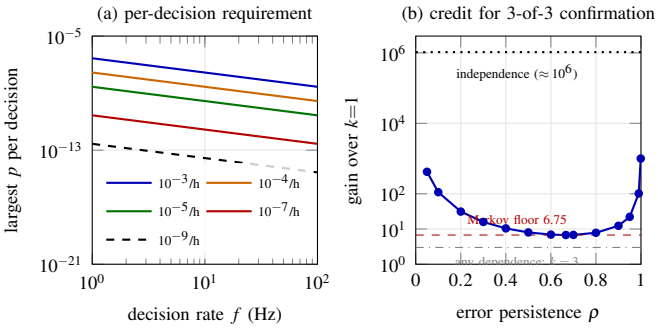


Fig. 4. The arithmetic of the per-flight-hour quantifier. (a) Largest per-decision hazard probability compatible with a per-hour target, $p^\star = \lambda/(3600 f)$ (Proposition 3(a)). At 30 Hz and 10^{-4}/h it is 9.3×10^{-10}. (b) What temporal confirmation buys at 30 Hz and 10^{-4}/h. Under independence the gain is about 10^6. Under a first-order Markov error process the gain is never below 6.75, whatever the persistence ρ, and under arbitrary stationary dependence it is only guaranteed to be $k = 3$ (Proposition 3(b)). Provenance: DERIVED (closed form).

B. H1: Where Guarantees Sit

Fig. 3 shows the adjudicated pilot by component and scope. Across both tiers, 64 % of the 44 perception guarantees quantify per input, against 19 % of the 32 control guarantees. For control, 56 % quantify per trajectory. H1 holds for perception and fails for control, as predicted, and the type plane should be read separately for the two. One qualification matters more than the headline. Among tier-A perception guarantees, those evaluated on vehicles, only 14 % are per input, because vehicle-facing perception work is dominated by closed-loop verification through surrogates. The literature has therefore moved beyond the per-input level for perception, but only through a lift whose assumption is statistical. H1 in its original form is too coarse. The refined statement is that per-input certificates dominate perception methods, while vehicle-facing perception results are per trajectory *conditional on* a $\langle \mathsf{pop_d}, \mathbb{P} \rangle$ surrogate assumption. Whether the full corpus confirms the tier-A share is the first thing it must settle, because the pilot over-samples closed-loop work by design.

TABLE IV

THE OBJECTIVE CORPUS, CODED BY TYPE AND HAZARD (N NATURAL, ADV ADVERSARIAL, B BOTH). †WORDING RESTS ON A SECONDARY SOURCE AND IS FLAGGED FOR CHECKING. GENERATED FROM `CORPUS/OBJECTIVES.CSV`.

ID	Document	Statement (paraphrased)	Type	Hazard	Target/FH
O01	JARUS SORA 2.5 Main Body (JAR-doc-25)	fewer than 1E-6 fatalities per flight hour	$\langle \mathsf{pop}_\mathsf{h}, \mathbb{P} \rangle$	N	1×10^{-6}
O02	JARUS SORA 2.5 Main Body (JAR-doc-25)	fewer than 1E-7 mid-air collisions per flight hour	$\langle \mathsf{pop}_\mathsf{h}, \mathbb{P} \rangle$	N	1×10^{-7}
O03	JARUS SORA 2.5 Annex E (JAR-doc-28)	probability of leaving the operational volume below 1E-3 per flight hour	$\langle \mathsf{pop}_\mathsf{h}, \mathbb{P} \rangle$	N	1×10^{-3}
O04	JARUS SORA 2.5 Annex E (JAR-doc-28)	probability of leaving the operational volume below 1E-4 per flight hour	$\langle \mathsf{pop}_\mathsf{h}, \mathbb{P} \rangle$	N	1×10^{-4}
O05	JARUS SORA 2.5 Annex E (JAR-doc-28)	no single failure leads to operation outside the ground risk buffer	$\langle \mathsf{op}, \forall \rangle$	N	–
O06	JARUS SORA 2.5 Main Body and Annex A (JAR-doc-26)†	contingency volume sized from GNSS accuracy and position-holding error and reaction and stopping distance; entry is an abnormal situation	$\langle \mathsf{op}, \forall \rangle$	N	–
O07	JARUS SORA 2.5 Annex E (JAR-doc-28)	30/300/3000/30000 flight hours for SAIL I-IV at 95 percent confidence	$\langle \mathsf{pop}_\mathsf{h}, \mathbb{P} \rangle$	N	–
O08	FAA AC 23.1309-1E	catastrophic failure condition below 1E-6 (Class I) to 1E-9 (Class IV) per flight hour	$\langle \mathsf{pop}_\mathsf{h}, \mathbb{P} \rangle$	N	1×10^{-9}
O09	FAA AC 23.1309-1E	no single failure results in a catastrophic failure condition	$\langle \mathsf{op}, \forall \rangle$	N	–
O10	EASA CS-25 AMC 25.1309†	catastrophic failure conditions of the order of 1E-9 per flight hour or less	$\langle \mathsf{pop}_\mathsf{h}, \mathbb{P} \rangle$	N	1×10^{-9}
O11	EASA AI Concept Paper Issue 02†	quantifiable generalisation bounds over the AI/ML constituent ODD	$\langle \mathsf{pop}_\mathsf{d}, \mathbb{P} \rangle$	N	–
O12	EASA AI Concept Paper Issue 02	generalisation-based error estimates are quantities on average	$\langle \mathsf{pop}_\mathsf{d}, \mathbb{E} \rangle$	N	–
O13	EASA AI Concept Paper Issue 02	data completeness and representativeness over the ODD as a prerequisite for generalisation bounds	$\langle \mathsf{pop}_\mathsf{d}, \mathbb{P} \rangle$	N	–
O14	EASA AI Concept Paper Issue 02†	stability of the model under perturbations and adverse conditions	$\langle \mathsf{in}, \forall \rangle$	B	–
O15	EASA AI Concept Paper Issue 02	monitoring to detect input data outside the AI/ML constituent ODD	$\langle \mathsf{op}, \forall \rangle$	N	–
O16	EASA AI Concept Paper Proposed Issue 03†	independent oversight system bounding Level 3 AI behaviour	$\langle \mathsf{op}, \forall \rangle$	N	–
O17	EASA MLEAP final report (2024)	generalisation relative to the ODD	$\langle \mathsf{pop}_\mathsf{d}, \mathbb{P} \rangle$	N	–
O18	EASA MLEAP final report (2024)	local robustness and stability of the trained model	$\langle \mathsf{in}, \forall \rangle$	B	–
O19	EUROCAE ED-324 / SAE ARP6983 (concept level)†	generalisation of the ML model	$\langle \mathsf{pop}_\mathsf{d}, \mathbb{P} \rangle$	N	–
O20	EUROCAE ED-324 / SAE ARP6983 (concept level)†	stability and robustness of the ML model	$\langle \mathsf{in}, \forall \rangle$	B	–
O21	ASTM F3269-21	safety monitor and recovery function bound the behaviour of a complex function on every execution	$\langle \mathsf{op}, \forall \rangle$	N	–
O22	RTCA DO-326A / EUROCAE ED-202A†	intentional unauthorised electronic interaction does not lead to an unacceptable threat condition	$\langle \mathsf{op}, \forall \rangle$	Adv	–

TABLE V

PILOT INTER-CODER RELIABILITY OVER 86 GUARANTEES: PERCENT AGREEMENT, COHEN'S κ, KRIPPENDORFF'S α WITH 95 % BOOTSTRAP INTERVAL, GWET'S AC1 AND PABAK. BOTH CODERS ARE INDEPENDENT LLM INSTANCES, SO THE TABLE MEASURES CODEBOOK CLARITY. GENERATED FROM `RESULTS/RELIABILITY_C1_C2.CSV`.

Field	%	κ	α [95 % CI]	AC1	PABAK
tier	92	0.84	0.84 [0.70, 0.93]	0.84	0.84
component	90	0.82	0.82 [0.71, 0.92]	0.88	0.87
scope	88	0.83	0.83 [0.72, 0.92]	0.85	0.84
modality	93	0.88	0.88 [0.78, 0.96]	0.92	0.91
type (scope×modality)	86	0.83	0.83 [0.74, 0.91]	0.85	0.85
perturbation	76	0.70	0.69 [0.57, 0.80]	0.72	0.72
assumption status	85	0.47	0.46 [0.21, 0.67]	0.83	0.80
evidence class	88	0.77	0.77 [0.62, 0.89]	0.86	0.84

C. H2: Are Lifts Accounted For?

Of the 54 pilot guarantees that use at least one lift, 4 % leave the lift's assumption unstated and 80 % state it without evidence. 4 % evidence it, and only 13 % carry that evidence's confidence into the final claim. In total 17 % account for the assumption as an obligation in the sense of Theorem 1. The rate differs by lift. The dynamics lift is the most common (25 papers) and is evidenced in 16 % of them: the plant model is stated and taken as given. The surrogate lift is evidenced in 45 % of 11 papers, the concentration lift in 35 % of 20, and the monitor lift in 0 % of 9. No pilot paper uses $\mathsf{L}_{\mathrm{sup}}$ explicitly (0), even though every per-operation guarantee needs it to reach a population objective. H2 is supported, and the pattern is sharper than predicted. The lift most often left as an assumption is the one regulators care most about: that deployment stays inside the certified set.

There is a trend. Among lifted guarantees from 2017–2021, 7 % account for their assumption (n = 27). From 2022–2026 the figure is 26 % (n = 27). Conformal methods under distribution shift, semi-probabilistic vision controllers, and learned perception contracts all make the surrogate's assumption a calibrated, confidence-carrying term. The full study must test whether this trend survives an unbiased sample.

D. H3: The Transfer Table

Table VI is the computed transfer table. Of its 102 cells, 8 are discharges, 66 need lifts, and 28 are category errors. Four patterns stand out.

No class discharges a per-flight-hour objective (T1). 15 of 17 classes reach T1, at least through $\mathsf{L}_{\mathrm{rate}}$, so no per-hour claim exists without exposure and dependence evidence. The cheapest classes (CP, PACB, OPREL) need only that one lift, which matches H3's prediction that statistical guarantees are the natural fit. Every per-input class needs three or four obligations. Proposition 2 is visible in the column: every T1 path carries a statistical obligation.

Statistical guarantees for perception are scarce. The pilot holds 20 guarantees with modality $\mathbb{P}$. 12 concern perception (27 % of perception guarantees), and only 5 of those are evaluated on vehicles. Estimation, mostly conformal trajectory prediction, is the exception at 83 %. Only 3 pilot papers perform the rate lift to a per-hour or per-demand system figure. H3 is supported.

Local robustness objectives (T6) are never discharged. EASA and MLEAP robustness refers to both natural and adversarial perturbations. Every per-input class covers one of them, so every cell carries O_{thr}. This is a modelling gap, not a proof gap: the certified sets of the literature were designed for an adversary that the objective does not single out.

Falsification and average-case analyses discharge nothing above their own type. The FALS row is all $\times$, and DTMC discharges only the average-error objective T3. This is the formal version of a familiar warning, that a failed search for counterexamples and an average case are not evidence for a worst-case or tail objective.

E. H4: Runtime Assurance

Of the 17 classes, 11 reach a per-operation objective (T4), and only RTA discharges one directly. This is H4's prediction, but Proposition 5 qualifies it. RTA carries 4 intrinsic obligations: monitor coverage, switching latency, verified recovery and independence. No pilot RTA paper evidences all four, and the evidenced share for the class is zero. For a perception-based monitor, coverage is a $\langle \mathsf{pop}_{\mathsf{d}}, \mathbb{P} \rangle$ statement about the monitor's learned input, so the architecture has moved the original gap onto its own monitor. H4 is supported in the form "RTA is the only direct bridge, and it relocates the deficit". It is not supported in the stronger form "RTA closes the gap". The independence obligation is where the two forms part. If the monitor reads the same perception channel it guards, the failures are common-cause and the lift collapses.

F. What the Per-Flight-Hour Quantifier Costs

Fig. 4 turns Propositions 3 and 4 into numbers. Panel (a) is the budget that a per-frame argument must meet. At 30 Hz it is 9.3×10^{-10} for 10^{-4}/FH, and 9.3×10^{-15} for 10^{-9}/FH. No certified-radius result in the corpus speaks to a quantity of this kind, because a radius bounds no probability. Panel (b) shows why the usual escape, temporal confirmation, needs O_{dep}. Without independence evidence, three-frame confirmation is guaranteed to buy between 3 and 6.75$\times$, not the $10^6\times$ that the independence calculation suggests.

VIII. Case Study: An Evidence Pack for the Running Example

We apply the generator to the composition certificate at its operating point: detector setting $\theta = 0.25$ s, $H = 2$ hops, re-anchoring horizon $T_c = 1$ s and measured local Lipschitz constant $L = 1.181$. Table VII shows the typed chain and the pack's obligations.

What the pack certifies: With the empirically calibrated rate $\gamma = 1.365$ m/s, the induced perturbation is 0.68 m, and the tube radius is $\rho = 2.22$ m. Taking the supremum rate over all calibration samples ($\gamma = 1.625$ m/s) gives 2.65 m. The position-holding term of the SORA contingency volume under the stated adversary is therefore certified at the 5, 10 and 20 m margins, and not at 2 m. Under the kinematic worst case it would be 24.4 m, and nothing would be certified. The pack writes the other contingency-volume terms (navigation accuracy, map error, reaction distance, contingency manoeuvre) as operator obligations, because the certificate says nothing about them.

What it does not certify: Of the pack's 6 obligations, 2 are open or assumed. The chain is a valid transfer path to $\langle \mathsf{op}, \forall \rangle$ (T4) and, through L_{sup}, to $\langle \mathsf{pop}_{\mathsf{d}}, \mathbb{P} \rangle$ (T2). It stops there: O_{dep} is open, so the pack supports no per-flight-hour claim, and it says so. The case also illustrates Corollary 1. The parent pays four obligations where two would buy a per-hour number, and in return it obtains the per-operation envelope that sizes a contingency volume. Neither route is wrong. The pack makes the choice visible.

Why this reframes a limitation: The parent bounds a proportion of missions, not a single flight, and a reviewer can read that as a weakness. In the type plane it is the correct level for an airworthiness argument. Every $\langle \mathsf{pop}_{\mathsf{h}}, \mathbb{P} \rangle$ objective is a population statement, and the only per-operation statement the parent could add is the $\langle \mathsf{op}, \forall \rangle$ corridor bound it already proves. This is a reframing, and we present it as one. It does not close the parent's interface-calibration limitation, which remains the pack's only partially evidenced obligation.

IX. Research Agenda

The empty and costly cells of Table VI define the agenda. We rank them by how many objectives they would discharge.

1) **Evidence for** O_{dep} (all T1 cells, 7 objectives). Per-hour objectives are reachable from every non-refuting class only through the rate lift. Temporal-dependence models of perception error, with confidence and validated on flight data, are the single most valuable missing piece. Proposition 3 shows that credit for confirmation turns on them.
2) **Reachable tubes as contingency volumes** (T4, 6 objectives). Q_{tr} tools could size SORA's position-holding term directly. Our search found no paper that does so; the evidence pack is a first instance. The missing step is O_{hor} under realistic re-anchoring, including denial.
3) **Calibrated surrogate coverage** (PIL to T2, T4). Perception-in-the-loop results become transferable once O_{cov} carries a confidence. Recent work does this for contracts and semi-probabilistic controllers; it should become the norm.
4) **Monitor coverage with confidence** (RTA and MON to T4). ASTM F3269 names coverage as a concern, and we found no pilot paper that evidences it with a confidence. Independence of the monitor's input must be argued, not assumed.
5) **Hazard models that match the objective** (all T6 cells). Robustness objectives name natural and adversarial perturbations together. Certified sets parametrized by the ODD, with an adversarial margin, would close the coverage obligation.

X. Related SoKs and Surveys

Prior systematizations catalogue one level. The SoK of Li et al. organizes certified robustness for deep networks

TABLE VI

THE COMPUTED TRANSFER TABLE. ROWS ARE GUARANTEE CLASSES WITH THEIR PILOT SIZE n AND MODAL TYPE. COLUMNS ARE OBJECTIVE TYPES: T1 PER FLIGHT HOUR $\langle \mathsf{pop_h}, \mathbb{P} \rangle$, T2 ODD GENERALIZATION $\langle \mathsf{pop_d}, \mathbb{P} \rangle$, T3 AVERAGE ERROR $\langle \mathsf{pop_d}, \mathbb{E} \rangle$, T4 PER OPERATION $\langle \mathsf{op}, \forall \rangle$ (N), T5 SECURITY THREAT CONDITION $\langle \mathsf{op}, \forall \rangle$ (ADV), T6 LOCAL ROBUSTNESS $\langle \mathsf{in}, \forall \rangle$ (B). **D**: DISCHARGE. L_k: LEAST PATH WITH k OBLIGATIONS, LIFTS LISTED (THR: HAZARD COVERAGE ONLY). $\times$: CATEGORY ERROR. LAST COLUMN: SHARE OF CLASS MEMBERS WHOSE LIFT ASSUMPTION IS EVIDENCED (SE OR SEC). GENERATED BY `CORPUS_ANALYSIS.PY`.

Class	n	type	T1 per FH	T2 ODD gen.	T3 average	T4 per op.	T5 security	T6 local rob.	ev. %
RS	6	$\langle \mathsf{in}, \forall_\alpha \rangle$	L_4 α,op,rate	L_3 α,op	L_3 α,op	L_4 α,dyn,hor	L_3 α,dyn,hor	L_2 α	0
BP	7	$\langle \mathsf{in}, \forall \rangle$	L_3 op,rate	L_2 op	L_2 op	L_3 dyn,hor	L_2 dyn,hor	L_1 thr	0
SMT	5	$\langle \mathsf{in}, \forall \rangle$	L_3 op,rate	L_2 op	L_2 op	L_3 dyn,hor	L_3 dyn,hor	L_1 thr	0
LIP	3	$\langle \mathsf{in}, \forall \rangle$	L_3 op,rate	L_2 op	L_2 op	L_3 dyn,hor	L_2 dyn,hor	L_1 thr	0
SEM	8	$\langle \mathsf{in}, \forall \rangle$	L_3 op,rate	L_2 op	L_2 op	L_3 dyn,hor	L_2 dyn,hor	L_1 thr	0
PIL	8	$\langle \mathsf{tr}, \forall \rangle$	L_3 hor,sup,rate	L_2 hor,sup	L_2 hor,sup	L_1 hor	L_2 hor	$\times$	62
DTMC	1	$\langle \mathsf{pop_d}, \mathbb{E} \rangle$	$\times$	$\times$	**D**	$\times$	$\times$	$\times$	100
CLR	14	$\langle \mathsf{tr}, \forall \rangle$	L_4 hor,sup,rate	L_3 hor,sup	L_3 hor,sup	L_2 hor	L_2 hor	$\times$	0
CRL	4	$\langle \mathsf{tr}, \forall_\alpha \rangle$	L_5 α,hor,sup,rate	L_4 α,hor,sup	L_4 α,hor,sup	L_3 α,hor	L_2 α,hor	$\times$	0
NCERT	2	$\langle \mathsf{tr}, \forall \rangle$	L_4 hor,sup,rate	L_3 hor,sup	L_3 hor,sup	L_2 hor	L_2 hor	$\times$	0
CP	5	$\langle \mathsf{op}, \mathbb{P} \rangle$	L_1 rate	**D**	**D**	$\times$	$\times$	$\times$	0
RTA	6	$\langle \mathsf{op}, \forall \rangle$	L_2 sup,rate	L_1 sup	L_1 sup	**D**	L_1 thr	$\times$	0
MON	3	$\langle \mathsf{in}, \forall \rangle$	L_2 op,rate	L_1 op	L_1 op	L_2 dyn,hor	L_3 dyn,hor	L_1 thr	0
FALS	3	$\langle \mathsf{op}, \exists \rangle$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	0
PACB	3	$\langle \mathsf{pop_d}, \mathbb{P} \rangle$	L_1 rate	**D**	**D**	$\times$	$\times$	$\times$	0
PROB	3	$\langle \mathsf{in}, \mathbb{P} \rangle$	L_3 op,rate	L_2 op	L_2 op	$\times$	$\times$	$\times$	33
OPREL	5	$\langle \mathsf{pop_d}, \mathbb{P} \rangle$	L_1 rate	**D**	**D**	$\times$	$\times$	$\times$	40

by verification and training technique, entirely at in [126]. Liu et al. survey verification algorithms [127]. Brunke et al. and Dawson et al. survey safe learning-based control and learned certificates [82], [128]. Hobbs et al. introduce runtime assurance [99]. Huang et al. and Mohseni et al. survey safety and trustworthiness of deep networks [129], [130]. Papernot et al. systematize security and privacy in ML [131]. Tambon et al. conduct a systematic review of how to certify ML-based safety-critical systems, and are our closest methodological precedent [132]. On the assurance side, Burton et al. name the semantic gap between specification and learned behaviour [133]. Ashmore et al. structure assurance across the ML lifecycle [134]. AMLAS provides safety-case patterns [135], and Assurance 2.0 and eliminative argumentation make residual doubt explicit [136], [137]. Seshia et al. set out the challenges of verified AI [138], and Bensalem et al. argue for combining statistical and formal guarantees [29]. None of these works types claims by what they quantify over, and none treats lifts as obligations whose deficit can be computed. That cross-level transfer analysis is our contribution. Our obligations map onto the assumption nodes and assurance claim points of goal structuring notation (GSN) [139], [140], so a transfer path can be exported as an argument fragment. Formally, the per-input level is a 2-safety hyperproperty [141], and the op/pop distinction is the distinction between path quantification and probabilistic quantification in probabilistic computation tree logic (PCTL) [142].

XI. LIMITATIONS AND THREATS TO VALIDITY

The corpus figures are a pilot. The pilot was seeded purposively and over-samples closed-loop and statistical work, so its distributions do not estimate the field. Its role is to exercise the protocol and to fix the codebook before the full study, which replaces every number reported here.

Both pilot coders were LLMs. Their agreement measures how unambiguous the codebook is, not how experts read the papers. The two coders also share a training distribution, which may inflate agreement. The full study uses two human coders and keeps the LLM as a measured third.

The calculus is ours. The verdicts depend on the edge set. We chose edges that correspond to arguments in the corpus, and we release the code so that a reader can add, remove or re-weight an edge and regenerate the table. Unit weights treat all obligations alike. A deficit weighted by evidence kind is a natural refinement.

TABLE VII
THE RUNNING EXAMPLE AS A TYPED TRANSFER PATH, FROM THE GENERATED EVIDENCE PACK (RESULTS/EVIDENCE_PACK.JSON). STATUS: E EVIDENCED, P PARTIALLY EVIDENCED, S STATED, A ASSUMED, O OPEN.

Step	Lift	Type	Obligation and evidence	St.
$\theta \mapsto \Delta$	Lemma	$\langle \mathsf{in}, \forall \rangle$	proved; cap measured over 15 392 hops	E
$\Delta \mapsto \delta$	calibration	$\langle \mathsf{in}, \forall \rangle$	γ transfers to deployment: 3 real flights, one airframe, hover	P
$\delta \mapsto \rho$	$\mathsf{L}_{\mathrm{dyn}}$	$\langle \mathsf{tr}, \forall \rangle$	$\mathsf{O}_{\mathrm{lip}}$: local L measured on the checkpoint	E
$\rho \mapsto$ corridor	$\mathsf{L}_{\mathrm{hor}}$	$\langle \mathsf{op}, \forall \rangle$	$\mathsf{O}_{\mathrm{hor}}$: re-anchoring every T_c; delay adversary only	A
corridor $\mapsto$ MCR	$\mathsf{L}_{\mathrm{sup}}$	$\langle \mathsf{pop}_{\mathsf{d}}, \mathbb{P} \rangle$	$\mathsf{O}_{\mathrm{odd}}$: mission distribution named, simulated	S
MCR $\mapsto$ per FH	$\mathsf{L}_{\mathrm{rate}}$	$\langle \mathsf{pop}_{\mathsf{h}}, \mathbb{P} \rangle$	$\mathsf{O}_{\mathrm{dep}}$: not claimed	O
hazard	—	—	$\mathsf{O}_{\mathrm{thr}}$: delay adversary below threshold; denial, insider and monitor-input attacks not covered	S

Class-level types hide within-class spread. A class takes its modal type. Members of other types are visible in Appendix D and in the released codes.

Regulatory drift. ED-324 and the EASA Proposed Issue 03 were moving at our cut-off, and $22 - 14$ objectives rest on secondary sources. We code concepts, not clause numbers, and flag every secondary objective.

Construct validity. "Accounted for" (SE or SEC) is a strict reading of H2. Under a looser reading that counts any stated assumption, the share rises to the sum of the S, SE and SEC shares. We report both, and the sensitivity analysis of the full study re-codes under each.

XII. CONCLUSION

Certified robustness and airworthiness assurance do not fail to meet because one is too weak. They quantify over different things. Once guarantees and objectives carry types, the gap becomes computable. Lifts bridge it, and each lift leaves an obligation behind. No chain of proofs reaches a probability objective without a statistical obligation. The cheapest route to a per-hour number skips the per-operation envelope that sizes a contingency volume. Runtime assurance moves the gap onto its monitor. In the pilot, the literature states most of its lift assumptions but evidences few of them, least of all the one regulators care about: that the aircraft stays where the certificate holds. The evidence generator shows the alternative. It lists every assumption as an obligation, and it keeps the boundary of the claim in the pack.

ETHICS CONSIDERATIONS

This work involves no human subjects and no personal data. It analyses published papers and public regulatory documents. Mapping guarantees to regulatory objectives could be misread as certification advice. We state throughout that it is not, and the evidence generator writes that disclaimer into every pack it emits. The LLM coding was run on bibliographic metadata and the coders' own knowledge of published papers; no unpublished material was processed.

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The codebook, the coded pilot corpus with both coders' codes and the adjudication log, the objective corpus, the calculus, the analysis scripts, the evidence generator and the runbook for the full study are released in the anonymised artifact. Every number in this paper is a macro generated from those files.

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APPENDIX A
PROOFS

Proof of Proposition 1. $\succeq$ is the reflexive–transitive closure of the free edges, so it is reflexive and transitive. Each free edge either increases the scope index or moves the modality from $\mathbb{P}$ to $\mathbb{E}$, and no edge reverses either, so the closure has no cycles and is antisymmetric. Every edge into $\langle \mathsf{pop_h}, \mathbb{P} \rangle$ is $\mathsf{L_{rate}}$, which is not free, so no free path reaches it from $\langle \mathsf{in}, \forall \rangle$. No edge decreases scope, so no path at all leads back. The same argument applies to $\langle \mathsf{op}, \forall \rangle$ and $\langle \mathsf{pop_d}, \mathbb{P} \rangle$, through $\mathsf{L_{sup}}$. □

Proof of Theorem 1. Each lift is a valid rule: if its premise and its obligations hold, its conclusion holds. Let F_0 be the event that g fails and F_{O} the event that the evidence for

O is invalid. Off $F_0 \cup \bigcup_{\mathsf{O}} F_{\mathsf{O}}$, induction along π shows that every intermediate conclusion holds. By the union bound, that event has probability at most δ_π. The error terms compose by substituting each rule's bound into the next. If the final type dominates τ_o, the objective follows by Definition 2. □

Proof of Proposition 2. The edges that change the modality from $\forall$ or $\forall_\alpha$ to $\mathbb{P}$ are L_{op}, L_{sup} and L_{mon}. They carry O_{prof}, O_{odd} and O_{mcov}, all statistical. L_α maps $\forall_\alpha$ to $\forall$, and every other edge preserves $\mathbb{P}$ or leaves it. Any path that ends at modality $\mathbb{P}$ from $\forall$ or $\forall_\alpha$ must therefore cross one of the three. The released code checks the claim exhaustively over all pairs of types. □

Proof of Proposition 3. (a) Let X_t indicate a hazardous decision. By linearity of expectation, $\mathbb{E}\sum_{t\le 3600f} X_t = 3600 f\, p$ for any joint law with marginals p. (b) Every hazardous run contains at least k errors, so the number of runs is at most the number of errors divided by k. Taking expectations gives the first bound. For the Markov chain, stationarity gives $\Pr(X_{t-1}=0, X_t=1) = \Pr(X_t=1, X_{t+1}=0) = p(1-\rho)$, and a run of length at least k starts at t with probability $p(1-\rho)\rho^{k-1}$. The function $(1-\rho)\rho^{k-1}$ is maximized at $\rho = (k-1)/k$, with value $(k-1)^{k-1}/k^k$. Under independence ($\rho = p$) the rate is $3600 f p^k (1-p)$. □

Proof of Proposition 4. (a) Under a Poisson process with rate λ, $\Pr(\text{no failure in } T) = e^{-\lambda T}$. The test rejects $\lambda \ge \lambda_0$ at level a if and only if $e^{-\lambda_0 T} \le a$. With $a = 0.05$, $T\lambda_0 = \ln 20 = 2.996$, so $T = 30, \ldots, 30\,000$ h give $\lambda_0 = 0.0999, \ldots, 9.99 \times 10^{-5}$. (b) With n independent Bernoulli frames, zero failures reject $p \ge p_0$ at level a when $(1-p_0)^n \le a$, that is, $n \ge \ln(1/a)/p_0$ to first order. With $p_0 = \lambda_0/(3600f)$ this is $n/(3600f) = \ln(1/a)/\lambda_0$ hours. □

Proof of Proposition 5. In the edge set, the only edges into $\langle \mathsf{op}, \forall \rangle$ are L_{hor} from $\langle \mathsf{tr}, \forall \rangle$ and the identity, so a direct discharge needs a guarantee already of that type. The Simplex theorem states safety on every execution in which the monitor flags each state outside the recoverable set within the switching latency, and the recovery controller is verified from every state it can be handed. These are O_{mcov}, O_{lat} and O_{rec}, and the monitor's input must not share the failure modes of the channel it guards, which is O_{ind}. When the monitor's input is a learned perception output, the event "the monitor flags every hazardous state" has a probability over deployed inputs, which is of type $\langle \mathsf{pop}_{\mathsf{d}}, \mathbb{P} \rangle$. □

Proof of Corollary 1. $\langle \mathsf{in}, \forall \rangle \xrightarrow{\mathsf{L}_{\text{op}}} \langle \mathsf{pop}_{\mathsf{d}}, \mathbb{P} \rangle \xrightarrow{\mathsf{L}_{\text{rate}}} \langle \mathsf{pop}_{\mathsf{h}}, \mathbb{P} \rangle$ carries $\{\mathsf{O}_{\text{prof}}, \mathsf{O}_{\text{dep}}\}$, and no path with fewer obligations exists, because every edge into $\langle \mathsf{pop}_{\mathsf{h}}, \mathbb{P} \rangle$ is L_{rate} and no free edge leaves $\langle \mathsf{in}, \forall \rangle$. Any path through $\langle \mathsf{op}, \forall \rangle$ must enter it by L_{hor} from $\langle \mathsf{tr}, \forall \rangle$, which is reached from $\langle \mathsf{in}, \forall \rangle$ only by L_{dyn}, and must leave it by L_{sup} and then L_{rate}. That gives $\{\mathsf{O}_{\text{lip}}, \mathsf{O}_{\text{hor}}, \mathsf{O}_{\text{odd}}, \mathsf{O}_{\text{dep}}\}$. The released code reproduces both paths. □

Appendix B
Search Strings

Block A (one query per phrase): "certified robustness", "certifiably robust", "formal verification", "reachability analysis", "runtime assurance", "simplex architecture", "conformal prediction", "PAC-Bayes", "statistical model checking", "scenario approach", "safety verification", "provable guarantee", "perception contract", "safety certificate", "barrier certificate", "operational profile". Block B: ("neural network" OR "deep learning" OR "learning-enabled" OR "machine learning" OR "reinforcement learning"). Block C: ("aircraft" OR "UAV" OR "UAS" OR "drone" OR "quadrotor" OR "autonomous vehicle" OR "autonomous driving" OR "avionics" OR "airworthiness" OR "collision avoidance" OR "safety-critical"). Years 2017–2026. Database-specific syntax is generated by `search_corpus.py`, which logs every request.

Appendix C
Codebook Decision Rules

1) Randomized-smoothing confidence is $\forall_\alpha$, never $\mathbb{P}$.
2) Falsification and counterexample search are $\exists$. A zero-failure sampling campaign whose confidence enters the claim is $\mathbb{P}$.
3) If a certificate over a domain is checked by sampling and the sampling enters the claim, add L_{conc}.
4) A competition report inherits the scope of its benchmark category.
5) Runtime-assurance architectures with a conditional safety theorem are $\langle \mathsf{op}, \forall \rangle$ with L_{mon}.
6) Between two candidate scopes, choose the smaller and flag it.
7) (v1.1, CLR-1) Initial-set reachability with no disturbance input has perturbation "none".
8) (v1.1, PIL-1) A result over a generator's latent space is "semantic". A result over an estimated perception-error set is "distribution".
9) (v1.1, COMP-1) A learned trajectory or intent predictor is "estimation".

Appendix D
Guarantee Classes of the Pilot

Appendix E
Acronyms

Class	Members
RS	[1], [53]–[57]
BP	[3], [13]–[16], [48], [49]
SMT	[2], [12], [47], [66], [67]
LIP	[50]–[52]
SEM	[58]–[64], [94]
PIL	[10], [87]–[93]
DTMC	[95]
CLR	[4], [5], [17], [69]–[79]
CRL	[83]–[86]
NCERT	[80], [81]
CP	[113]–[117]
RTA	[27], [68], [96]–[98], [100]
MON	[101]–[103]
FALS	[105]–[107]
PACB	[109]–[111]
PROB	[119]–[121]
OPREL	[24], [25], [122]–[124]

AC1	Gwet's first-order agreement coefficient
ACAS Xu	airborne collision avoidance system for unmanned aircraft
AINNCS	AI and neural-network control systems (ARCH-COMP category)
AMC	acceptable means of compliance
BP	bound propagation
CP	conformal prediction
DTMC	discrete-time Markov chain
EASA	European Union Aviation Safety Agency
EKF	extended Kalman filter
EUROCAE	European Organisation for Civil Aviation Equipment
FAA	Federal Aviation Administration
FH	flight hour
GSN	goal structuring notation
IBP	interval bound propagation
JARUS	Joint Authorities for Rulemaking of Unmanned Systems
LLM	large language model
MCR	mission-completion rate
ML / NN	machine learning / neural network
MLEAP	Machine Learning Application Approval (EASA project)
NNCS	neural-network control system
OD / ODD	operational domain / operational design domain
PABAK	prevalence-adjusted bias-adjusted kappa
PAC	probably approximately correct
PCTL	probabilistic computation tree logic
PRISMA	Preferred Reporting Items for Systematic reviews and Meta-Analyses
RS	randomized smoothing
RTA	run-time assurance
SAIL	specific assurance and integrity level
SMT	satisfiability modulo theories
SoK	systematization of knowledge
SORA	Specific Operations Risk Assessment
STL	signal temporal logic
TLS	target level of safety
UAS / UAV	unmanned aircraft system / unmanned aerial vehicle
VNN-COMP	International Verification of Neural Networks Competition